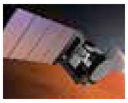


Methane removal
ESA scientists outline the possible mechanisms by which methane is outgassed and removed. **https://dgit.in/jan20-15**

The oxygen mystery
Curiosity has posed another puzzle for astronomers, one where oxygen behaves in a radical new way. **https://dgit.in/jan20-16**

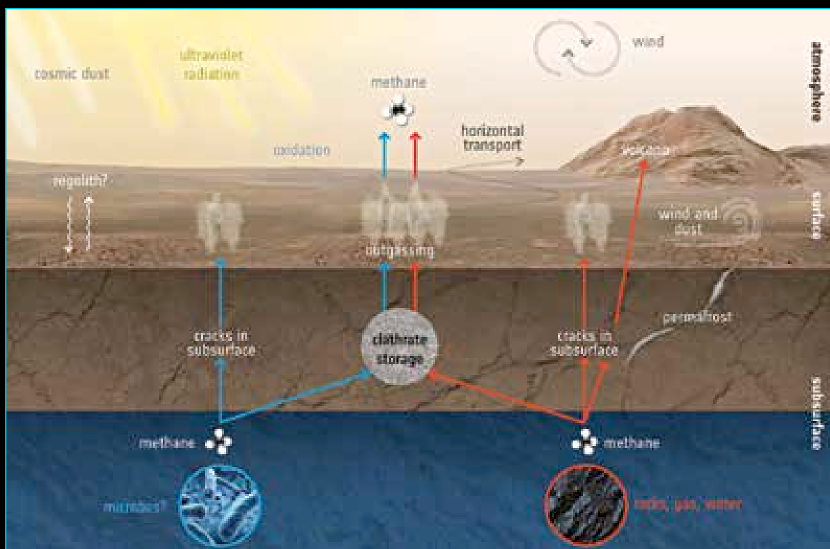
Over the years, Curiosity has regularly measured the abundance of methane in the atmosphere. In 2018, after six Earth or three Mars years of study, Scientists released data measurements over time. The outgassing increase in the summer months, and decrease in the winter. Also, the methane levels appear to increase during the night, when the temperatures are lowest, and decrease during the day. From these trends, scientists can guess that the concentration of methane in the Martian atmosphere is tied to the exposure to sunlight. The outgassing is believed to increase in the summer months, and during the daytime, as the natural cracks in the regolith expand due to the rise in temperature. While announcing the seasonal variation of methane, Chris Webster of NASA's Jet Propulsion Laboratory in California said, "This is the first time we've seen something repeatable in the methane story, so it offers us a handle in understanding it."

One of discrepancies in the measurements was that the orbiters were not observing the same levels of methane as the rover. Now methane degrades on exposure to ultraviolet radiation. Curiosity was busy observing rocks in the day, and only measured methane levels at night. The Curiosity team did make an exception with a daytime measurement, but this was not during a spike. In any case, it was believed that the methane degrades in the atmosphere, which is why the orbiters were not detecting it. However, the methane was disappearing from the atmosphere too rapidly for UV radiation to be an adequate explanation. Apart from a mysterious origin, there is an as yet unknown process that rapidly removes methane from the atmosphere, as soon as it is dumped there.

Researchers from Aarhus University showed that the methane could bind with the minerals known to exist on the surface of Mars such as plagioclase and basalt. According to these researchers, the Methane was simply being absorbed by the ground again after outgassing. The mechanism is more effective than UV radiation at removing methane from the atmosphere.

ESA scientists have identified several possible natural processes by which methane could be removed from the atmosphere. The idea is that the methane seeps through the surface of the planet, finding its way to the rocks below again. The strong winds on Mars could dilute the concentrations of methane from local outgassing, making it difficult for orbiters and Earth based instruments to detect.

Over the course of the three Martian year study, Curiosity found another baffling measurement. During the summer months, the carbon dioxide is released from the polar caps, increasing the air pressure of the atmosphere. In winter, the air pressure reduces as the carbon dioxide freezes again. The concentration of nitrogen and argon predictably track the concentration of carbon dioxide. However, oxygen does not do this on Mars. The amount of oxygen in the atmosphere increases by 30 percent in the summer months, and reduces to expected levels in the winter months. This spike of oxygen in summer months is beyond any known chemistry. At least on some occasions, the fluctuations of the oxygen and methane seem to be in tandem. This makes some scientists wonder if the same processes are responsible for the variation in both oxygen and methane. This is a brand new enigma, one that the Curiosity team announced in November 2019. Melissa Trainer, a planetary scientist at NASA who led the research said, "This is the first time where we're seeing this interesting behavior over multiple years. We don't totally understand it. For me, this is an open call to all the smart people out there who are interested in this: See what you can come up with."



Source: ESA

The creation and destruction of methane on Mars

Unfortunately, the Mars 2020 rover and the Rosalind Franklin rover both do not have payloads on board meant for the detection of methane, so the source of the methane is likely to remain a mystery for the next decade. The Trace Gas Orbiter is however, studying the concenteration of Methane in detail. NASA, ESA, CNSA and Roscosmos have proposed sample return missions to the Red Planet in the 2030s, which may finally answer the question as labs on Earth will be able to determine if the methane is biological in origin or not, based on the particular isotopes of the elements in methane. d